LESSON 01 / 13

## Integration reverses differentiation on an interval

A function $F$ is an antiderivative of $f$ on an interval $I$ when $F' = f$ there. We write

$$\int f(x)\,dx = F(x) + C.$$

If $F_1' = F_2' = f$ on $I$, then $(F_1 - F_2)' = 0$. The Mean Value Theorem therefore gives $F_1 - F_2 = C$: all antiderivatives on that interval form one vertical family.

The differential $dx$ identifies the variable. Thus $\int 2x\,dx = x^2 + C$, whereas $\int 2x\,dt = 2xt + C$ when $x$ is independent of $t$.

**WORKED EXAMPLE**

**Reverse the power, exponential and trigonometric rules**

Linearity allows term-by-term integration:

$$\int x^3\,dx = \tfrac{x^4}{4} + C, \qquad \int x^5\,dx = \tfrac{x^6}{6} + C,$$

$$\int (3x^2 + 2x + 1)\,dx = x^3 + x^2 + x + C, \qquad \int (3x^2 + 2x)\,dx = x^3 + x^2 + C.$$

Reverse the chain rule and include the inner scale:

$$\int e^{2x}\,dx = \tfrac{1}{2}e^{2x} + C, \qquad \int e^{3x}\,dx = \tfrac{1}{3}e^{3x} + C,$$

$$\int \sin(2x)\,dx = -\tfrac{1}{2}\cos(2x) + C.$$

Every answer can be checked immediately by differentiating it.

## The basic antiderivative library comes from derivatives

| Integrand | Antiderivative | Condition |
|---|---|---|
| x^n | x^(n+1)/(n+1) | n is not -1 |
| 1 / x | ln\|x\| | x is nonzero |
| e^x | e^x | all real x |
| sin x | -cos x | all real x |
| cos x | sin x | all real x |
| sec^2 x | tan x | cos x is nonzero |
| csc^2 x | -cot x | sin x is nonzero |
| sec x tan x | sec x | cos x is nonzero |

READ THE FIGURE
*The exceptional power n=-1 produces a logarithm. Trigonometric formulas retain the domains on which their derivatives exist.*

## An initial value selects one member of the family

WORKED EXAMPLE
**Use the condition after finding the general antiderivative**
*Suppose* $F'(x) = 3x^2 + 2x + 1$ *and* $F(0) = 5$. *Integration gives*

$$F(x) = x^3 + x^2 + x + C.$$

*Now* $F(0) = C = 5$, *so the required function is*

$$F(x) = x^3 + x^2 + x + 5.$$

*The initial condition does not change the derivative; it fixes the vertical shift.*

COMMON TRAP
**One constant per connected interval**
*On* $(0, \infty)$, $\int x^{-1}dx = \ln x + C$; *on* $(-\infty, 0)$, *it is* $\ln(-x) + C$. *The compact formula* $\ln|x| + C$ *works on either interval separately. A single constant need not describe both branches simultaneously.*

LESSON 02 / 13

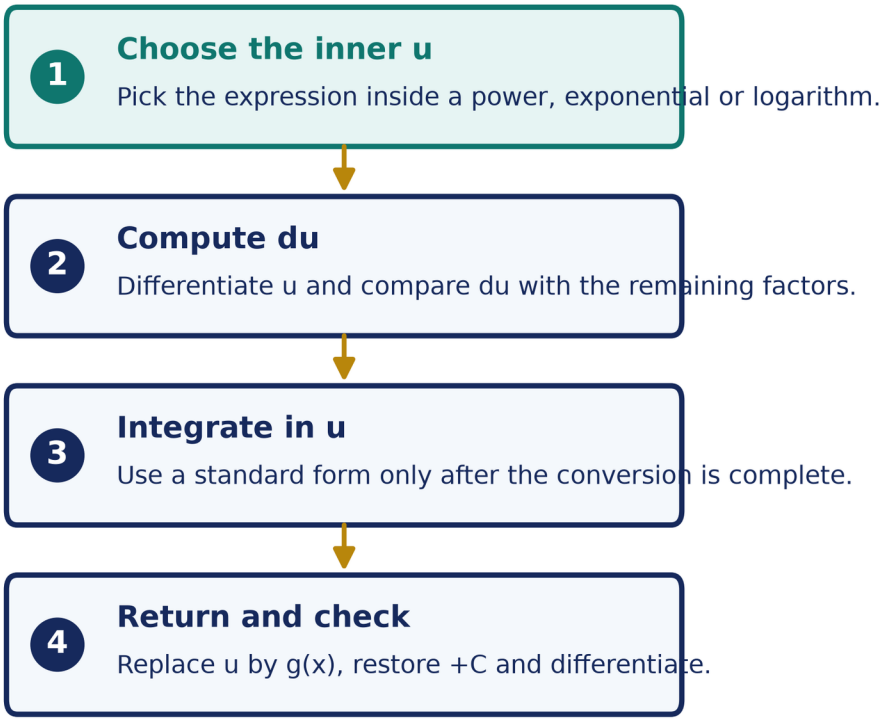
## Look for an inner expression and its derivative

From $(F(g(x)))' = F'(g(x))g'(x)$, we obtain

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du, \qquad u = g(x), \quad du = g'(x)dx.$$

Substitution is not a guess followed by hope: the factor accompanying $dx$ must account for $du$, up to a constant that can be moved outside the integral.

### A substitution audit prevents a missing scale factor




READ THE FIGURE
A constant multiplier can be repaired. A missing nonconstant factor usually means that another method or another substitution is needed.

WORKED EXAMPLE

**The derivative factor is already present**

For

$$I = \int 2x(x^2+1)^5\,dx,$$

put $u = x^2+1$, so $du = 2x\,dx$. Then $I = \int u^5 du = (x^2+1)^6/6 + C$.

Likewise, with $u = x^2+3$,

$$\int 2x(x^2+3)^4\,dx = \frac{(x^2+3)^5}{5} + C.$$

WORKED EXAMPLE

**A constant factor of one-half is enough**

For $\int xe^{x^2}dx$, put $u = x^2$, so $du = 2x\,dx$ and $x\,dx = du/2$. Hence

$$\int xe^{x^2}dx = \tfrac{1}{2}\int e^u du = \tfrac{1}{2}e^{x^2} + C.$$

The same antiderivative is used in the source's definite questions P4 and Q7; their limits and numerical evaluation belong to Chapter 21.

## Linear inner functions have a predictable scale

If $F' = f$ and $a \neq 0$, then

$$\int f(ax+b)\,dx = \tfrac{1}{a}F(ax+b) + C.$$

For example,

$$\int \cos(3x-1)dx = \tfrac{1}{3}\sin(3x-1) + C, \qquad \int (2x+1)^4 dx = \frac{(2x+1)^5}{10} + C.$$

COMMON TRAP

**Do not invent the derivative factor**

For $\int x(x^2+1)^5 dx$, the missing factor is only a constant 2, so substitution works with $x\,dx = du/2$. For $\int (x^2+1)^5 dx$, the needed factor $2x$ is absent and cannot be created as a constant multiplier.

LESSON 03 / 13

## A numerator equal to the denominator's derivative gives a logarithm

Where $g(x) \neq 0$,

$$\int \frac{g'(x)}{g(x)}\,dx = \ln|g(x)| + C.$$

The absolute value is essential on negative branches. With $u = 3x + 2$,

$$\int \frac{dx}{3x+2} = \frac{1}{3}\ln|3x+2| + C.$$

Similarly,

$$\int \frac{2x+2}{x^2+2x+2}dx = \ln(x^2+2x+2) + C,$$

where the quadratic is always positive, so absolute-value signs are optional.

**WORKED EXAMPLE**

**Tangent and cotangent are logarithmic substitutions**

Since $\tan x = \sin x/\cos x$, put $u = \cos x$ on an interval where $\cos x \neq 0$:

$$\int \tan x\,dx = -\int \frac{du}{u} = -\ln|\cos x| + C = \ln|\sec x| + C.$$

In the same way,

$$\int \cot x\,dx = \ln|\sin x| + C.$$

## Four standard trigonometric logs and their domains

| Integral | Antiderivative | Valid away from |
|---|---|---|
| tan x | ln\|sec x\| | zeros of cos x |
| cot x | ln\|sin x\| | zeros of sin x |
| sec x | ln\|sec x + tan x\| | zeros of cos x |
| csc x | ln\|csc x - cot x\| | zeros of sin x |

**READ THE FIGURE**

Each formula is proved by differentiation on one connected interval. Constants can change when a different interval is chosen.

## Secant and cosecant use a multiplier that creates $u'/u$

For secant, let $u = \sec x + \tan x$. Then

$$\frac{du}{dx} = \sec x\tan x + \sec^2 x = \sec x(\sec x + \tan x) = \sec x\, u.$$

Therefore $u'/u = \sec x$, giving the table's formula. For cosecant, $u = \csc x - \cot x$ satisfies $u' = \csc x\, u$.

**WORKED EXAMPLE**

**Check a scaled quotient rather than memorising its coefficient**

$$\int \frac{6x}{3x^2+5}dx = \ln(3x^2+5) + C,$$

because the numerator is exactly the derivative of the denominator. In contrast,

$$\int \frac{x}{3x^2+5}dx = \frac{1}{6}\ln(3x^2+5) + C.$$

**COMMON TRAP**

**A logarithm does not remove a singularity**

The formulas above are valid only between consecutive excluded points. Never integrate through a zero of the original denominator as though $\ln|g(x)|$ made the integrand finite there.

LESSON 04 / 13

## Simplify before choosing a technique

*Many integrals become elementary after division, expansion or a trigonometric identity. Preparation is part of the calculation, not an optional cosmetic step:*

$$\int \frac{x+1}{\sqrt{x}}dx = \int (x^{1/2} + x^{-1/2})dx = \frac{2}{3}x^{3/2} + 2x^{1/2} + C \qquad (x > 0).$$

**WORKED EXAMPLE**

***A direct power substitution***

*For* $\int \sin^3 x \cos x\, dx$, *put* $u = \sin x$, $du = \cos x\, dx$:

$$\int \sin^3 x \cos x\, dx = \int u^3 du = \frac{\sin^4 x}{4} + C.$$

## Odd and even trigonometric powers need different preparation

*For an odd power of sine, reserve one sine factor for* $du = -\sin x\, dx$ *and convert the remaining even power with* $\sin^2 x = 1 - \cos^2 x$:

$$\int \sin^3 x\, dx = \int (1 - \cos^2 x)\sin x\, dx = -\cos x + \frac{\cos^3 x}{3} + C.$$

*For even powers, halve the angle:*

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

*Consequently,*

$$\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C, \qquad \int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C.$$

*These antiderivatives also underpin the source's definite questions P10, Q24 and Q36; evaluation over limits is deferred to Chapter 21.*

## Choose a trigonometric preparation from the powers

| Pattern | Preparation | Likely next step |
|---|---|---|
| sin^n x cos x | u = sin x | power rule |
| cos^n x sin x | u = cos x | power rule |
| odd sine power | save sin x dx | u = cos x |
| odd cosine power | save cos x dx | u = sin x |
| even powers | half-angle identities | linear trigonometric terms |
| product sin x cos x | u = sin x or double angle | equivalent answers |

**READ THE FIGURE**

*The aim is to create either a complete du or a sum of standard forms. Do not mix the odd-power and even-power strategies midway.*

**WORKED EXAMPLE**

***Different-looking answers can all be correct***

*Three routes to* $I = \int \sin x \cos x \, dx$ *give*

$$I = \tfrac{1}{2}\sin^2 x + C, \qquad I = -\tfrac{1}{2}\cos^2 x + C, \qquad I = -\tfrac{1}{4}\cos 2x + C.$$

*Their derivatives are all* $\sin x \cos x$*. Moreover,*

$$-\tfrac{1}{2}\cos^2 x = \tfrac{1}{2}\sin^2 x - \tfrac{1}{2}, \qquad -\tfrac{1}{4}\cos 2x = \tfrac{1}{2}\sin^2 x - \tfrac{1}{4}.$$

*The apparent discrepancies are constants, not errors.*

**COMMON TRAP**

***Keep the original domain after cancellation***

*Algebraic simplification may expose a removable factor, but it does not restore points where the original integrand was undefined. State the simplified antiderivative only on intervals of the original domain.*

LESSON 05 / 13

## Recognise a sum of squares or a square-root difference

For $a>0$, direct differentiation gives

$$\int \frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+C, \qquad \int \frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+C,$$

the second on $-a<x<a$. Thus

$$\int \frac{dx}{x^2+4}=\frac{1}{2}\tan^{-1}\frac{x}{2}+C, \qquad \int \frac{dx}{\sqrt{4-x^2}}=\sin^{-1}\frac{x}{2}+C.$$

The source's definite Q10–Q12 use these same antiderivatives before their limits are applied in Chapter 21.

**WORKED EXAMPLE**

**Complete the square before using arctangent**

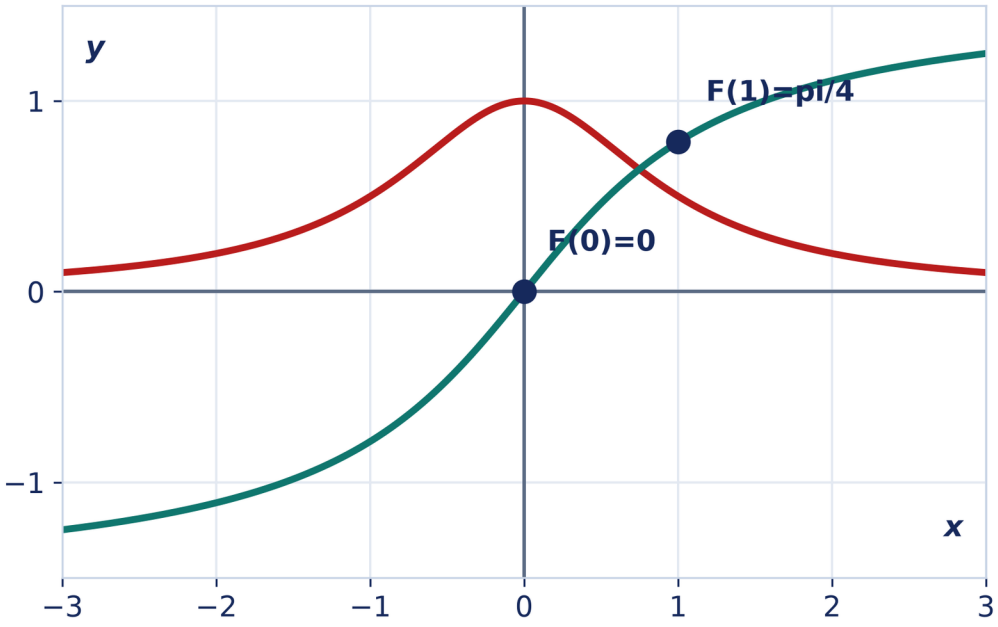
For SS,

$$x^2+2x+2=(x+1)^2+1.$$

With $u=x+1$ and $du=dx$,

$$\int \frac{dx}{x^2+2x+2}=\int \frac{du}{u^2+1}=\tan^{-1}(x+1)+C.$$

**Arctangent has derivative 1/(1+x^2)**




READ THE FIGURE

The teal curve is F(x)=arctan(x); the red curve is f(x)=1/(1+x^2). The slope of F at x=0 is 1 and at x=1 is 1/2: red gives height, teal gives the antiderivative.

## Scaling must be handled before quoting the formula

**WORKED EXAMPLE**

**Extract the coefficient of x squared**

$$\int \frac{dx}{4x^2+9} = \frac{1}{4}\int \frac{dx}{x^2+9/4} = \frac{1}{6}\tan^{-1}\frac{2x}{3} + C.$$

*Differentiating the last expression returns* $1/(4x^2+9)$*, including the factor* $1/6$*.*

**COMMON TRAP**

**Inverse notation is not reciprocal notation**

*Here* $\tan^{-1}u$ *and* $\sin^{-1}u$ *mean* $\arctan u$ *and* $\arcsin u$*, not* $1/\tan u$ *or* $1/\sin u$*. Also, the arcsin formula requires* $a>0$ *and* $|x|<a$*; it is not valid at an arbitrary complex or outside-domain input.*

LESSON 06 / 13

## Logarithms and inverse trigonometry come from the denominator's shape

For $a > 0$, on an interval avoiding singularities,

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C,$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C.$$

The first is Q21. Its partial-fraction derivation is

$$\frac{1}{x^2 - a^2} = \frac{1}{2a} \left( \frac{1}{x - a} - \frac{1}{x + a} \right).$$

### Three radical denominators have logarithmic antiderivatives

| Integrand | Antiderivative | Domain interval |
|---|---|---|
| 1/sqrt(x^2+a^2) | ln(x+sqrt(x^2+a^2)) | all real x |
| 1/sqrt(x^2-a^2) | ln\|x+sqrt(x^2-a^2)\| | x>a or x<-a |
| 1/sqrt(a^2-x^2) | arcsin(x/a) | -a<x<a |

READ THE FIGURE
The first two formulas can be checked by differentiating the radical and simplifying. On a disconnected domain, treat each interval separately.

## A radical in the numerator may need a trigonometric substitution

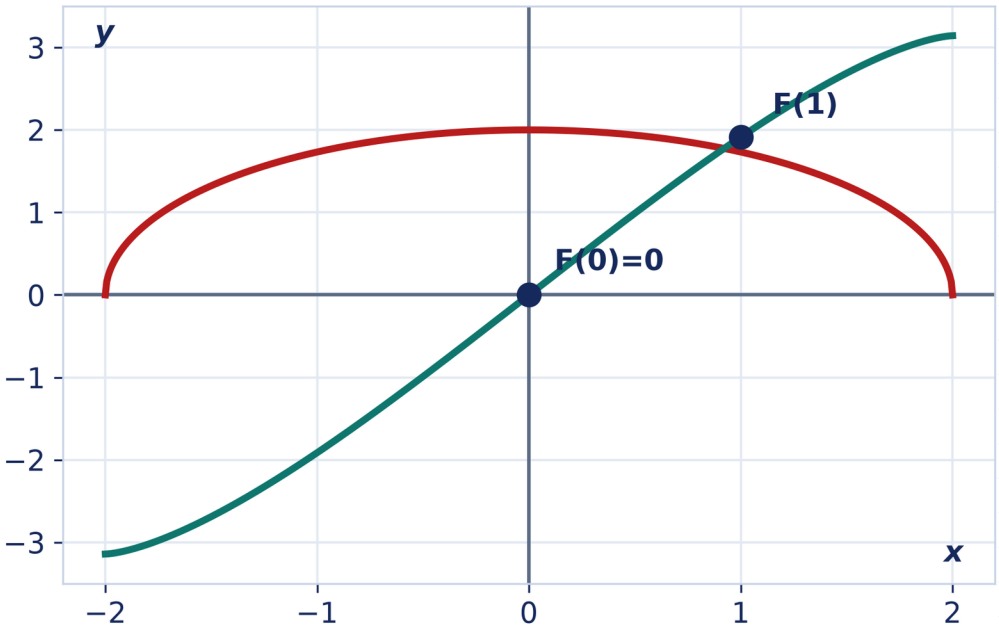
For $-a < x < a$, put $x = a \sin\theta$ with $-\pi/2 < \theta < \pi/2$. Then $dx = a\cos\theta\, d\theta$ and $\sqrt{a^2 - x^2} = a\cos\theta$, so

$$\int \sqrt{a^2 - x^2}\, dx = a^2 \int \cos^2\theta\, d\theta = \frac{a^2}{2}(\theta + \sin\theta\cos\theta) + C.$$

Returning to $x$ gives

$$\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\frac{x}{a} + C.$$

## The semicircle is the derivative of its antiderivative




READ THE FIGURE

The red semicircle is f(x)=sqrt(4-x^2); the teal curve is F=x sqrt(4-x^2)/2+2 arcsin(x/2). Differentiating the plotted F returns the red height.

COMMON TRAP

**Do not confuse the two difference-of-squares orders**

The answers for $1/(x^2 - a^2)$ and $1/(a^2 - x^2)$ differ by more than a casually moved sign. Their logarithmic ratios are different. Differentiation is faster and safer than trying to memorise the numerator order without a check.

LESSON 07 / 13

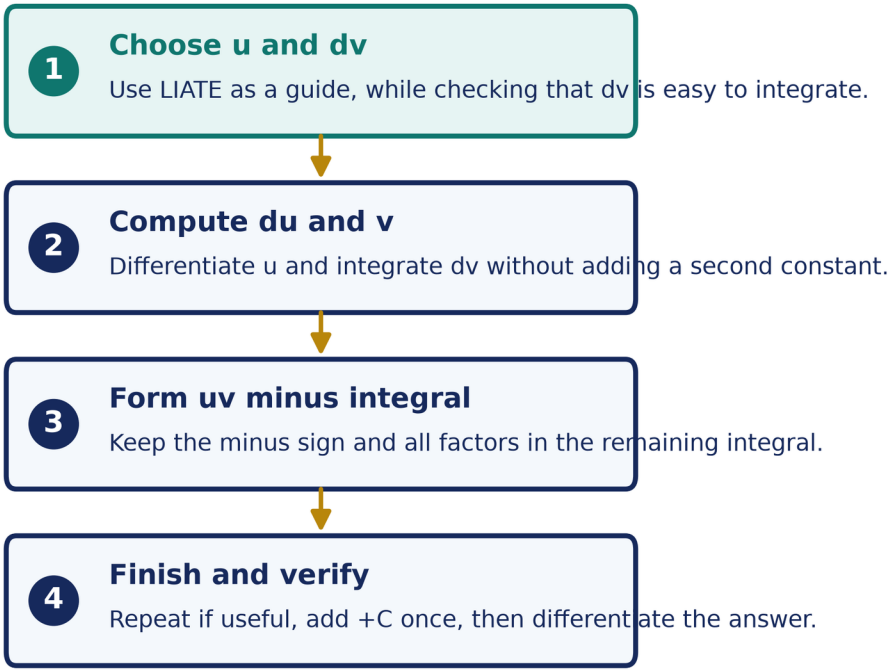
## Integrate the product rule instead of guessing a product rule

From $(uv)' = u'v + uv'$, we obtain

$$\int u\,dv = uv - \int v\,du.$$

A successful choice makes $\int v\,du$ simpler than $\int u\,dv$. LIATE is a useful default: choose $u$ from Logarithm, Inverse trigonometric, Algebraic, Trigonometric, Exponential, in that order.

### By parts transfers difficulty from u to du




READ THE FIGURE
The method succeeds when du is simpler or the remaining integral has a known recurrence. Reversing a good choice often makes the problem harder.

WORKED EXAMPLE

**One application removes the polynomial factor**

For S6 and Q14, choose $u = x$ and $dv = e^x dx$. Then $du = dx$ and $v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx = e^x(x-1) + C.$$

WORKED EXAMPLE

**A lone logarithm still has a product structure**

For S7 and Q15 on $x > 0$, take $u = \ln x$ and $dv = dx$:

$$\int \ln x\,dx = x\ln x - \int x \cdot \tfrac{1}{x}\,dx = x\ln x - x + C.$$

## Repeated parts can be organised as a polynomial recurrence

For P6 and Q16,

$$\int x^2e^x dx = x^2e^x - 2\int xe^x dx = x^2e^x - 2e^x(x-1) + C.$$

Therefore

$$\int x^2e^x dx = e^x(x^2 - 2x + 2) + C.$$

Differentiation gives $e^x(x^2 - 2x + 2) + e^x(2x - 2) = x^2e^x$.

**WORKED EXAMPLE**

***Inverse trigonometric functions also benefit from dv=dx***

$$\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}dx = x\tan^{-1} x - \tfrac{1}{2}\ln(1+x^2) + C.$$

**COMMON TRAP**

***Do not add independent constants at every by-parts step***

A constant chosen for $v$ can be absorbed into the remaining calculation. Write one final $+C$ for the complete antiderivative, and check that no intermediate constant has changed the intended recurrence.

LESSON 08 / 13

## A product with sine may need only one application

For the antiderivative used in Q17,

$$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x + C.$$

Its definite evaluation on $[0,\pi]$ is a Chapter 21 application of the Fundamental Theorem.

## Exponential-trigonometric products can return to the original integral

WORKED EXAMPLE

**Solve for the cyclic integral instead of abandoning it**

Let $I = \int e^x \sin x\,dx$ and $J = \int e^x \cos x\,dx$. By parts,

$$I = e^x \sin x - J, \qquad J = e^x \cos x + I.$$

Substitute the second equation into the first:

$$I = e^x \sin x - e^x \cos x - I.$$

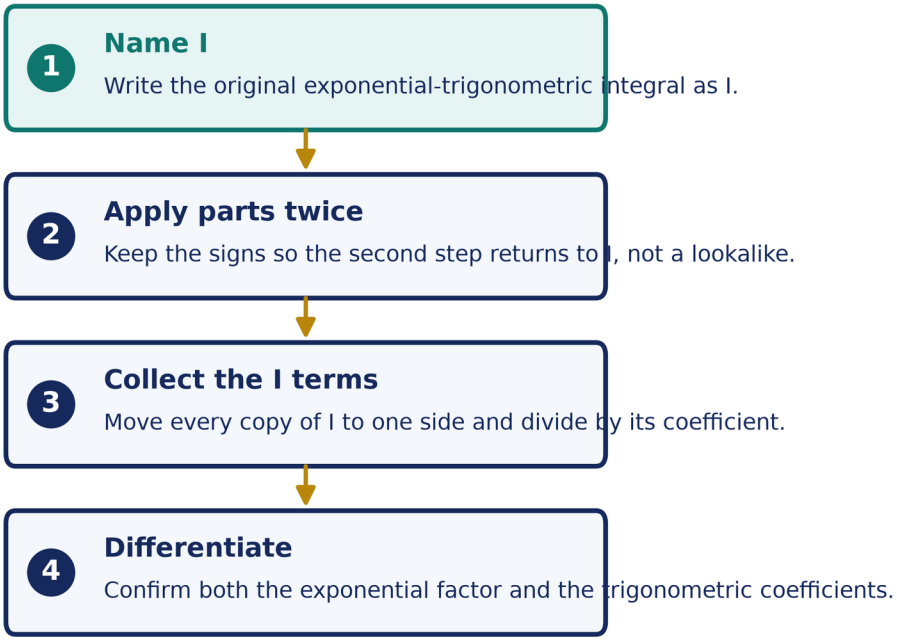
Thus P7 and Q18 have the antiderivative

$$\int e^x \sin x\,dx = \tfrac{1}{2}e^x(\sin x - \cos x) + C.$$

Differentiation gives

$$\tfrac{1}{2}e^x(\sin x - \cos x) + \tfrac{1}{2}e^x(\cos x + \sin x) = e^x \sin x.$$

## A cyclic by-parts calculation ends with algebra




READ THE FIGURE
*For e^x sin x, the calculation gives 2I=e^x(sin x-cos x). The factor 1/2 comes from collecting the returned integral.*

## An algebraic factor can hide a substitution after parts

WORKED EXAMPLE

**Combine by parts with a logarithmic substitution**

For $I=\int x\tan^{-1}x\,dx$, choose $u=\tan^{-1}x$ and $dv=x\,dx$:

$$I=\frac{x^2}{2}\tan^{-1}x-\frac{1}{2}\int\frac{x^2}{1+x^2}dx.$$

Since $x^2/(1+x^2)=1-1/(1+x^2)$,

$$I=\frac{x^2}{2}\tan^{-1}x-\frac{x}{2}+\frac{1}{2}\tan^{-1}x+C=\frac{x^2+1}{2}\tan^{-1}x-\frac{x}{2}+C.$$

COMMON TRAP

**The returned integral must have exactly the same sign and coefficient**

*If two applications produce $I=\text{terms}-I$, then $2I=\text{terms}$. Writing $I=\text{terms}$ at that stage loses the essential factor $1/2$. If the result is $I=I$, no information has been gained; change the arrangement.*

LESSON 09 / 13

## First make the rational function proper

A rational function is proper when the numerator has smaller degree than the denominator. If not, divide first:

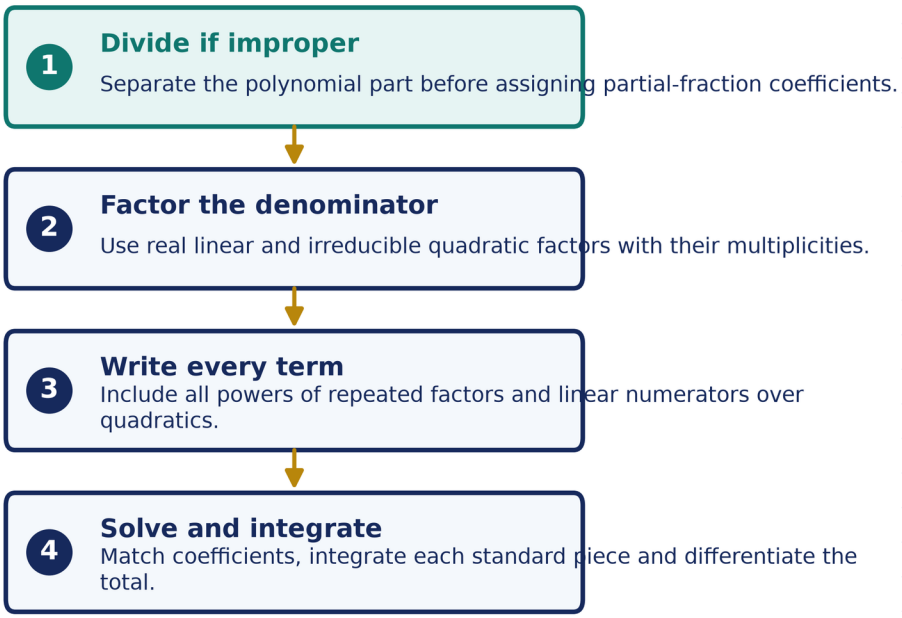
$$\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}.$$

Therefore

$$\int \frac{x^2+1}{x+1}dx = \frac{x^2}{2} - x + 2\ln|x+1| + C.$$

Only then factor the denominator and decompose the proper remainder.

### A rational function has a reliable preparation order




READ THE FIGURE
Most partial-fraction errors occur before integration: no prior division, an omitted repeated power or an incomplete numerator over a quadratic.

## Distinct linear factors produce logarithms

**WORKED EXAMPLE**

**S8 and Q19: decompose before integrating**

Since

$$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2},$$

we obtain

$$\int \frac{dx}{(x+1)(x+2)} = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

The definite Q20 applies limits to this antiderivative in Chapter 21; no singularity may lie inside the interval used.

## A repeated linear factor needs every power up to its multiplicity

For a factor $(x-a)^k$, include

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$

**WORKED EXAMPLE**

**Repeated factor: one logarithm and one reciprocal power**

Write

$$\frac{2x+3}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}.$$

Then $2x+3 = A(x+1) + B$, so $A = 2$ and $B = 1$. Hence

$$\int \frac{2x+3}{(x+1)^2}dx = 2\ln|x+1| - \frac{1}{x+1} + C.$$

**COMMON TRAP**

**A single A/(x-a) is incomplete for a repeated factor**

Omitting $(x-a)^{-2}, \ldots, (x-a)^{-k}$ usually makes the coefficient equations impossible—or, worse, produces an answer that matches at one point but has the wrong derivative. Differentiate the complete result on an allowed interval.

LESSON 10 / 13

## Split the numerator into derivative plus constant

Let $D = x^2 + px + q$ be irreducible, so $D > 0$ after completing the square. For a numerator $Ax + B$, write

$$Ax + B = \alpha(2x + p) + \beta, \qquad \alpha = \frac{A}{2}, \quad \beta = B - \frac{Ap}{2}.$$

Then

$$\int \frac{Ax + B}{D} dx = \alpha \ln D + \beta \int \frac{dx}{D} + C.$$

The remaining integral becomes arctangent after completing the square.

### Match the decomposition to the real denominator factor

| Factor | Required term | Integrated shape |
|---|---|---|
| x-a | A/(x-a) | logarithm |
| (x-a)^k | terms through power k | log and reciprocal powers |
| irreducible quadratic | (Ax+B)/quadratic | log plus arctangent |
| improper rational | polynomial division first | polynomial plus proper part |

READ THE FIGURE

The numerator over an irreducible quadratic must be linear. Splitting it into alpha D' + beta determines exactly when the logarithm term stands alone.

**WORKED EXAMPLE**

***P8 correction: the remainder beta is zero***

*For*

$$I = \int \frac{x+1}{x^2+2x+2}dx,$$

*the denominator's derivative is* $2x+2 = 2(x+1)$. *Thus*

$$x+1 = \tfrac{1}{2}(2x+2) + 0,$$

*so the complete answer is*

$$I = \tfrac{1}{2}\ln(x^2+2x+2) + C.$$

*The source adds* $\tfrac{1}{2}\tan^{-1}(x+1)$. *That extra term is incorrect: its derivative is*

$$\tfrac{1}{2}\cdot\frac{1}{(x+1)^2+1} = \frac{1}{2(x^2+2x+2)} \neq 0.$$

*Differentiating the corrected answer returns exactly* $(x+1)/(x^2+2x+2)$.

**WORKED EXAMPLE**

***A nonzero remainder produces both standard shapes***

*For* $D = x^2+2x+5 = (x+1)^2+4$,

$$3x+5 = \tfrac{3}{2}(2x+2) + 2.$$

*Therefore*

$$\int \frac{3x+5}{x^2+2x+5}dx = \tfrac{3}{2}\ln(x^2+2x+5) + 2\int \frac{dx}{(x+1)^2+4}.$$

*The last integral is* $\tfrac{1}{2}\tan^{-1}((x+1)/2)$, *hence*

$$I = \tfrac{3}{2}\ln(x^2+2x+5) + \tan^{-1}\tfrac{x+1}{2} + C.$$

**COMMON TRAP**

***Check the coefficient split before integrating either piece***

*A quick derivative check of the proposed identity* $Ax+B = \alpha D' + \beta$ *catches a missing remainder immediately. The logarithm and arctangent terms are not interchangeable decorations; their coefficients are forced by that identity.*

LESSON 11 / 13

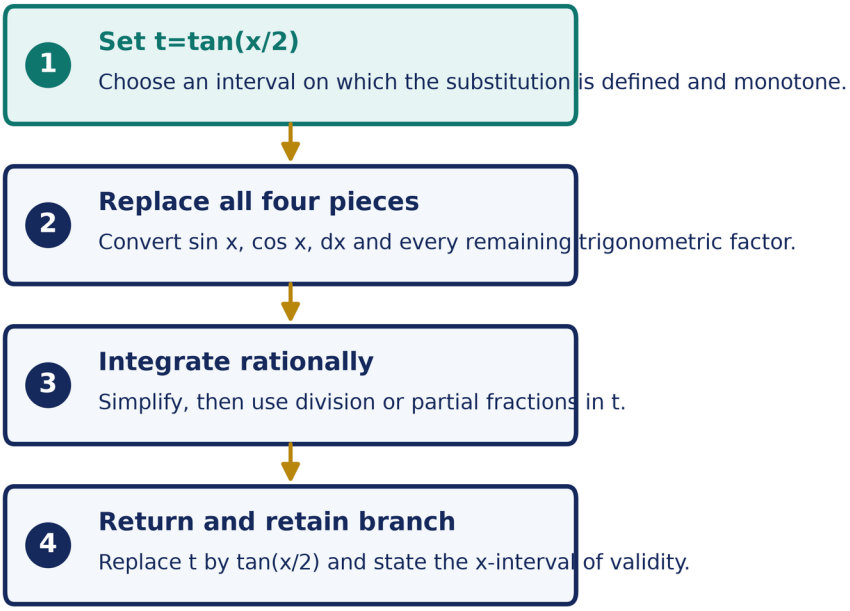
## Convert sine and cosine into rational functions of one variable

For a rational expression in $\sin x$ and $\cos x$, the universal substitution is

$$t = \tan\frac{x}{2}, \qquad \sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad dx = \frac{2\,dt}{1+t^2}.$$

It is powerful but algebra-heavy, so first check for a simpler identity, symmetry or direct substitution.

### The half-angle substitution is a complete change of variable




READ THE FIGURE

The factor dx=2dt/(1+t^2) is essential. Omitting it can coincidentally produce a familiar expression but cannot produce the correct antiderivative.

**WORKED EXAMPLE**

***The antiderivative behind S13 and Q35***

For

$$I=\int \frac{dx}{2+\cos x},$$

the denominator becomes

$$2+\cos x=2+\frac{1-t^2}{1+t^2}=\frac{3+t^2}{1+t^2}.$$

Including $dx=2dt/(1+t^2)$ gives

$$I=\int \frac{2}{3+t^2}dt=\frac{2}{\sqrt{3}}\tan^{-1}\frac{t}{\sqrt{3}}+C.$$

Thus, on $-\pi<x<\pi$,

$$I=\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{\tan(x/2)}{\sqrt{3}}\right)+C.$$

The definite value on $[0,\pi/2]$ is completed in Chapter 21.

## Another route can give an equivalent-looking answer

For $\int dx/(1+\sin x)$, the same substitution gives

$$\int \frac{2\,dt}{(1+t)^2}=-\frac{2}{1+t}+C=-\frac{2}{1+\tan(x/2)}+C.$$

On an allowed branch this differs from $\tan x-\sec x$ only by a constant, and both differentiate to $1/(1+\sin x)$.

**COMMON TRAP**

***The substitution has branches***

$t=\tan(x/2)$ is undefined at $x=(2k+1)\pi$. The displayed arctangent antiderivative is not automatically one continuous antiderivative across those points. Use it on a single branch; definite problems also require an interval free of the original integrand's singularities.

LESSON 12 / 13

## Diagnose the structure before calculating

*The fastest method is usually the one exposed by preparation. Do not begin with a favourite technique and force the integrand to fit it.*

### A compact method-selection map

| Visible structure | First method | Decisive check |
|---|---|---|
| polynomial or basic library | reverse derivative | power and scale |
| f(g(x)) g'(x) | u=g(x) | du is complete |
| g'(x)/g(x) | log substitution | g is nonzero |
| sum or difference of squares | standard form | complete square first |
| polynomial times e^x or trig | by parts | du gets simpler |
| log or inverse trig factor | by parts with dv=dx | new integral is easier |
| rational function | division and partial fractions | factor multiplicities |
| rational sin and cos | half-angle substitution | branch and dx factor |

**READ THE FIGURE**
*Preparation can move an integrand from one row to another. If a calculation grows without creating a standard form, stop and reassess the structure.*

## Three short choices show the difference

**WORKED EXAMPLE**

***Substitution, parts and preparation are not interchangeable***

For $\int xe^{x^2}dx$, the factor $x$ helps create $du = 2x\,dx$:

$$\int xe^{x^2}dx = \tfrac{1}{2}e^{x^2} + C.$$

For $\int xe^{x}dx$, there is no inner $x^2$; by parts gives

$$\int xe^{x}dx = e^{x}(x-1) + C.$$

For $\int (x^{1/2} + x^{-1/2})dx$, no advanced method is needed:

$$\int (x^{1/2} + x^{-1/2})dx = \tfrac{2}{3}x^{3/2} + 2x^{1/2} + C.$$

## Verification is a mathematical step, not a courtesy

A complete check has four parts:

1. Differentiate the proposed antiderivative and simplify to the **original** integrand.
2. Confirm the inner scale, signs and partial-fraction coefficients.
3. State an interval on which both integrand and antiderivative are defined.
4. If two answers look different, test whether their difference is constant on that interval.

For example, the three answers to $\int \sin x\cos x\,dx$ in `jee-04.md` differ by $0$, $-1/2$ or $-1/4$ when written relative to $\tfrac{1}{2}\sin^2 x$. Their equality up to $C$ is therefore proved, not inferred from similar graphs.

**COMMON TRAP**

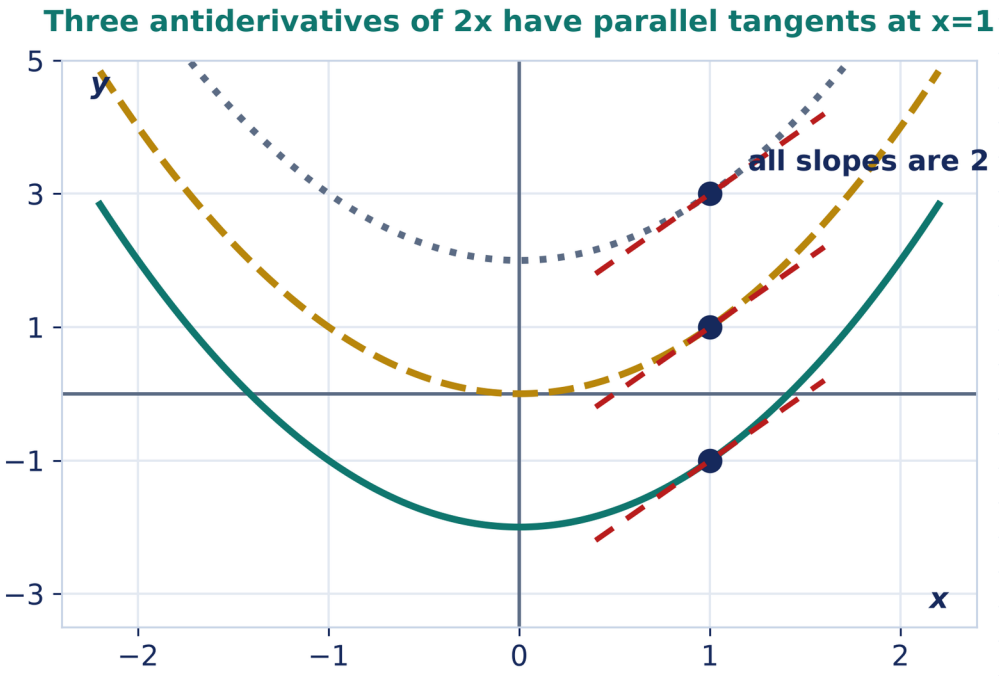
***A value check at one point is not enough***

Two different functions can agree at one point. Differentiation checks the whole relationship at once. Conversely, if $G' = f$ but the domain has been ignored, the calculation is still incomplete: singularities and disconnected branches remain part of the answer.

LESSON 13 / 13

An antiderivative is not merely an algebraic string. These six sheets display vertical families, domains, slopes, equivalent forms and substitution branches. Every curve is generated from an editable expression and independently differentiated in the verification suite.

## Lab 1 — The constant moves a curve without changing its slopes



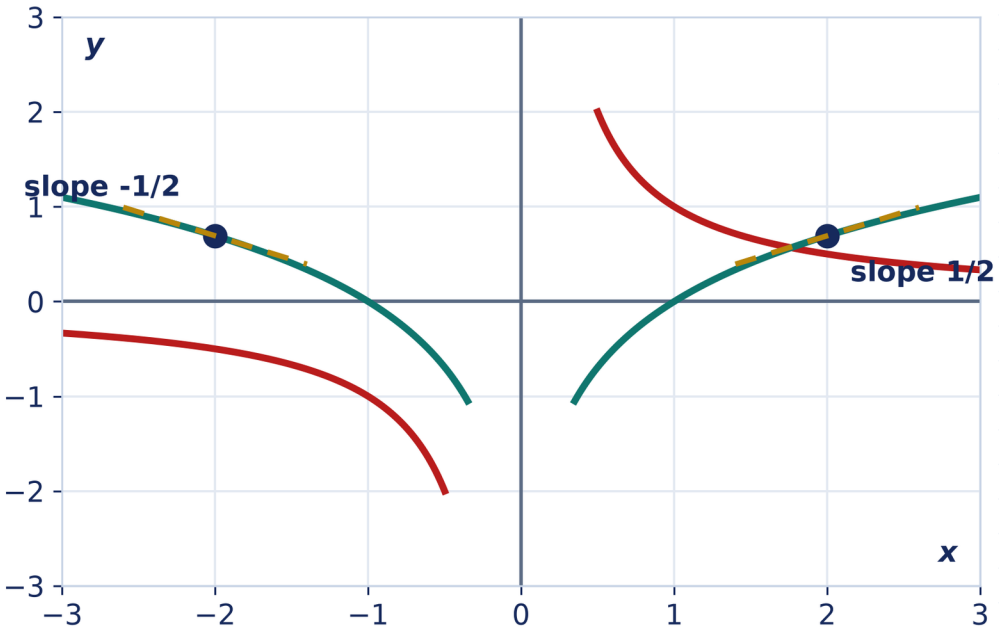

READ THE FIGURE

The curves x^2-2, x^2 and x^2+2 differ only by vertical shifts. Their red tangent segments at x=1 are parallel because each derivative is 2x=2.

If $F' = f$, then $(F + C)' = f$. Conversely, on one interval, any two antiderivatives have zero derivative for their difference and therefore differ by a constant. An initial condition selects exactly one of these vertical translates.

## Lab 2 — Logarithm branches and their slopes

### ln|x| differentiates to 1/x on both sides of zero



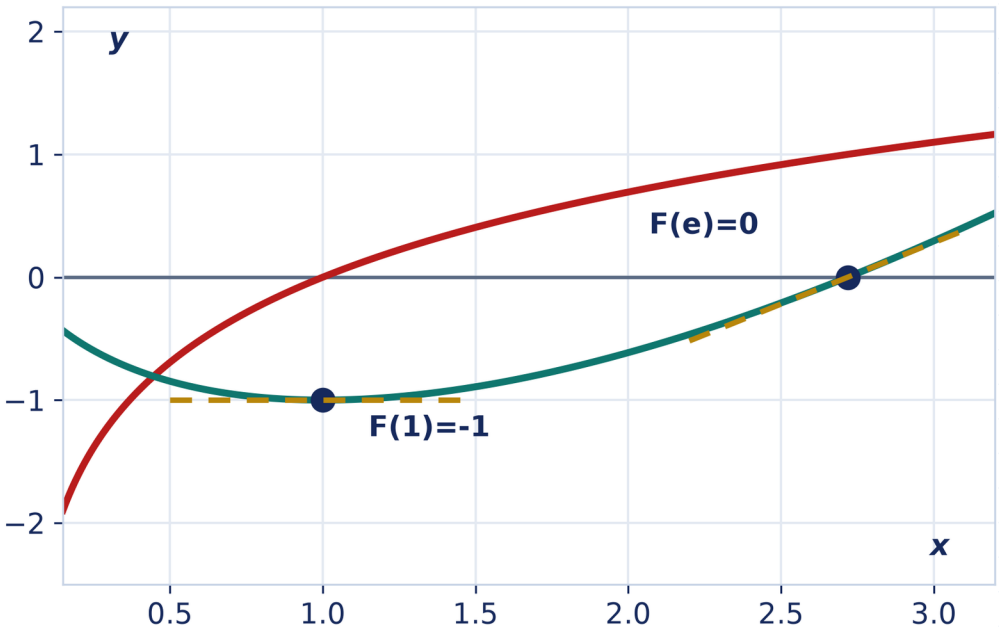

READ THE FIGURE

Red branches are f(x)=1/x; teal branches are F(x)=ln|x|. The gold tangent slopes at x=-2 and x=2 are -1/2 and 1/2. No curve crosses the excluded line x=0.

For $x<0$, $\ln|x| = \ln(-x)$, whose derivative is $(-1)/(-x) = 1/x$. The absolute value therefore handles both signs. The two disconnected branches may carry different constants even though the shorthand writes one $C$.

## Lab 3 — By parts produces a function whose slope is ln x

### The antiderivative x ln x - x has tangent slope ln x

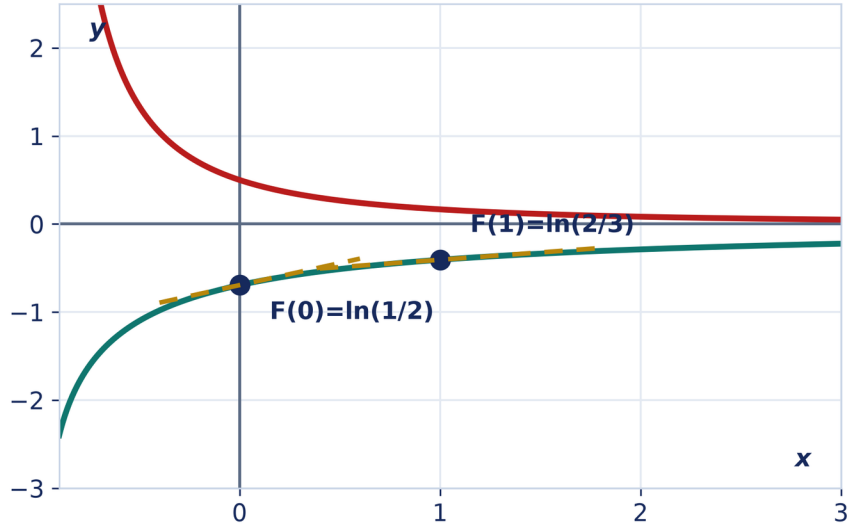



**READ THE FIGURE**
*Red is f(x)=ln x and teal is F(x)=x ln x−x. The gold tangent at x=1 is horizontal because ln 1=0; at x=e its slope is 1 because ln e=1.*

*The product-rule calculation $\int \ln x\,dx = x\ln x - \int 1\,dx$ is visible in the slopes. Notice that the red curve gives the **slope** of the teal curve, not its height. This distinction is the geometric meaning of $F' = f$.*

## Lab 4 — Partial fractions make a logarithm increase

### A positive partial-fraction integrand makes its antiderivative increase



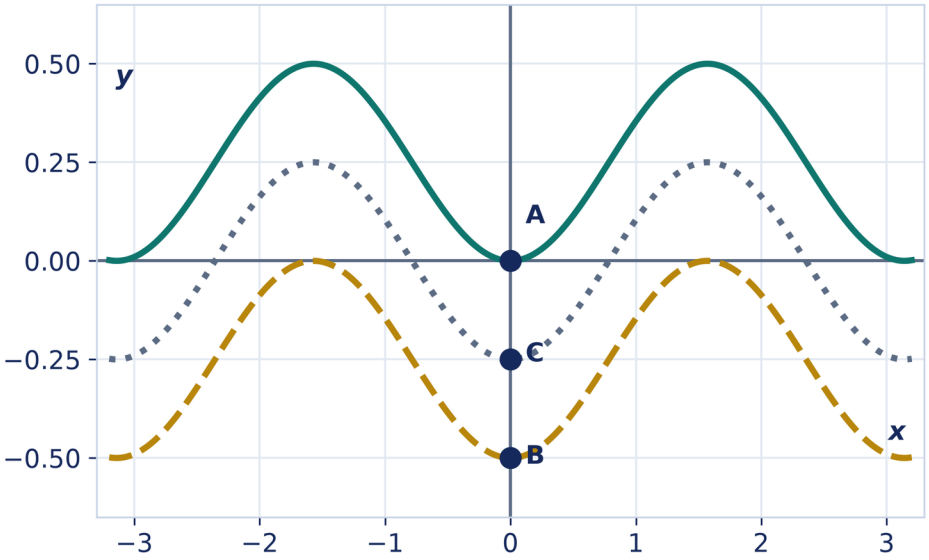

READ THE FIGURE

On x>-1, red is f=1/((x+1)(x+2)) and teal is F=ln|(x+1)/(x+2)|. Gold tangent slopes are f(0)=1/2 and f(1)=1/6, so F increases with decreasing slope.

The decomposition $f = 1/(x+1) - 1/(x+2)$ explains why the antiderivative is a difference of logarithms. The graph uses only the interval $x > -1$; the excluded points $-2$ and $-1$ divide the full domain into separate branches.

## Lab 5 — Equivalent answers are vertical translates

### Three correct antiderivatives of sin x cos x have the same shape

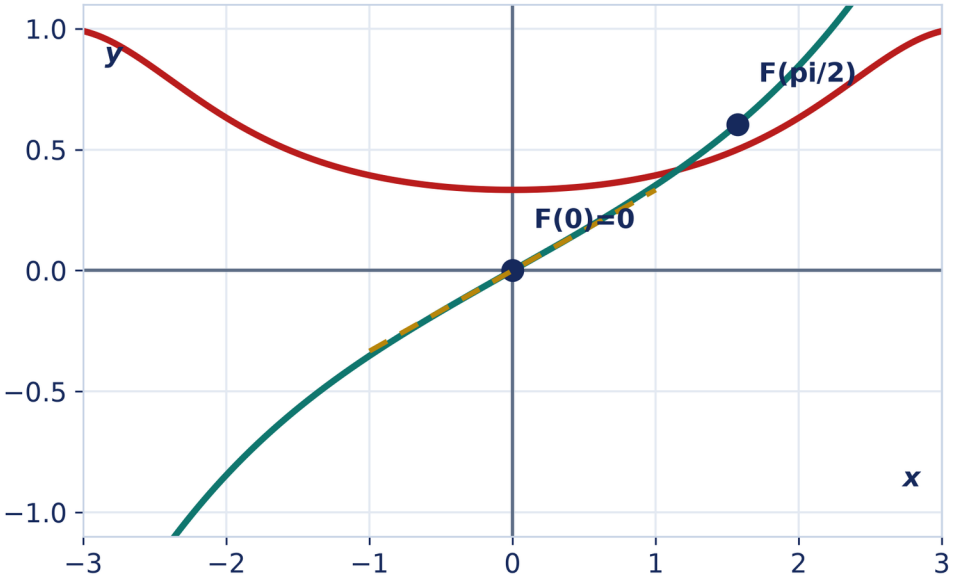



**READ THE FIGURE**
A=sin^2(x)/2, B=-cos^2(x)/2 and C=-cos(2x)/4. Their vertical differences are A-B=1/2 and A-C=1/4, so all three derivatives equal sin x cos x.

The curves are not approximations of one another: their vertical separations are exactly constant. This is why an answer can look different from a textbook answer and still be completely correct. Differentiate first; then compare constants if necessary.

## Lab 6 — Half-angle antiderivatives have branches

### The half-angle formula differentiates to 1/(2+cos x) on (-pi,pi)




READ THE FIGURE
Red is f=1/(2+cos x); teal is the half-angle antiderivative F. The gold tangent at zero has slope f(0)=1/3. At pi/2 the slope of F is f(pi/2)=1/2, not the displayed height.

On $-\pi < x < \pi$, $t = \tan(x/2)$ is continuous and strictly increasing, and differentiating the plotted $F$ returns the positive red integrand. At $x = \pm\pi$, the substitution itself fails. A continuous antiderivative on a larger interval must be constructed branch by branch with matching constants.